Short-term bus load forecasting based on combined feature selection and GRU-Attention Model

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***Abstract*—Short-term bus load forecasting is of great importance for power system dispatch and operation. In order to improve the accuracy of short-term bus load forecasting, a bus load forecasting method based on two feature selection algorithms and a gated cyclic unit with attention mechanism is proposed. This method firstly uses the pearson correlation coefficient method and the distance correlation coefficient method to obtain the correlation coefficient between weather feature quantities and bus load, thereby establishing a comprehensive correlation coefficient. Then, strongly correlated weather feature quantities are obtained based on the size of the comprehensive correlation coefficient, and input them into the gated cycle unit with attention mechanism model along with the historical bus load, and output the final prediction result. Through the verification of the actual bus load data validation, the method proposed in this paper achieves higher accuracy than the conventional forecasting methods.**

Keywords- bus load forecasting; feature selection; correlation coefficient; attention mechanism

I. INTRODUCTION

Accurate load forecasting is of great importance for the safe and stable operation of the grid. Depending on the object of the forecast, the load can be divided into system load and bus load. In particular, bus load can be defined as the sum of the end loads supplied to a relatively small area by the main transformer of a substation [1]. Compared to the system load, the bus load is of a smaller order of magnitude, the load components of each busbar are different, uncertainty is more pronounced [2] and forecasting is relatively difficult. Short-term bus load forecasting can span from a few hours to a few days [3], and its improved forecasting accuracy can provide more accurate data to support the development of generation plans and future time-of-use tariffs in the electricity market.

In recent years, scholars have carried out extensive research on bus load forecasting methods, which can be broadly classified into the categories of mathematical statistics and artificial intelligence. Typical mathematical statistics methods include linear extrapolation based on similar days [4], and curve extrapolation [5]. However, these methods often suffer from low forecasting accuracy, especially when dealing with nonlinear prediction tasks under large amounts of data. In contrast, artificial intelligence methods have the potential to provide more accurate predictions, and typically involve techniques such as support vector machine [6,7], gradient lifting tree algorithm [8], artificial neural network [9], and others. For instance, Ref. [10] utilizes hierarchical clustering to select similar days and establish a bus load forecasting model based on extreme learning machine. In Ref. [11], the equivalent load model of the bus is constructed using non-adjustable power generation, and the equivalent load is predicted using support vector machine. Ref. [12] proposes a method for predicting bus load based on a combination of autoregressive moving average and support vector functions. Moreover, artificial intelligence methods also include cyclic neural network based on deep learning, deep confidence network, and more. For instance, Ref. [13] uses phase space reconstruction to project bus load history data into phase space, and then input the results into a deep confidence network to obtain the predicted value of the bus load. In Ref. [14], bus load is decomposed based on empirical mode decomposition, and a long and short-term memory neural network is used to predict bus load values. Despite the progress made in bus load prediction accuracy, much research has yet to fully analyze the impact of meteorological factors and the significance of feature selection in the prediction process. Additionally, the influence degree of different input feature quantities on the final predicted value is often ignored when using deep learning networks to forecast.

In this paper, a short-term bus load forecasting method combining feature selection and Gated Recurrent Uint-Attention Mechanism (GRU-Attention) is proposed to forecast the bus load for the coming day. Before forecasting, the meteorological characteristics are pre-processed, the correlation coefficients between each meteorological characteristic and the bus load are calculated using Pearson Correlation Coefficient (PCC) and Distance Correlation Coefficient (DCC) respectively, then the features with high correlation are screened out according to the formula of comprehensive correlation coefficient defined in this paper, and combined with historical bus load data as input to the GRU-Attention forecasting model. Finally, the actual bus load on the high voltage side of a 220kV substation in China is used as an example to verify the superiority and effectiveness of the short-term bus load prediction method proposed in this paper.

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II. Feature selection algorithm

A. PCC

The PCC is a correlation coefficient that is primarily used to measure the linear correlation of two variables and can be easily obtained for characteristics with high correlation. Its calculation formula is shown in equation (1).

$$\rho_{PCC}=\frac{\sum xy-\frac{\sum x\sum y}{n}}{\sqrt{\left(\sum x^2-\frac{\left(\sum x\right)^2}{n}\right)\left(\sum y^2-\frac{\left(\sum y\right)^2}{n}\right)}} \tag{1}$$

where ρ_{PCC} is the correlation coefficient between busbar load and influencing factors, y is bus load, x is weather factor related to bus load, and n is the number of data contained in x or y .

B. DCC

The DCC can compensate for the fact that the PCC can only measure linear relationships between variables, and it can indicate non-linear relationships between variables. The distance correlation coefficient is calculated as shown below [15]:

$$\rho_{DCC}=\frac{d\operatorname{cov}(x,y)}{\sqrt{d\operatorname{cov}(x,x)d\operatorname{cov}(y,y)}} \tag{2}$$

where ρ_{DCC} is the calculated value of the distance correlation coefficient.

C. Integrated correlation coefficient

Based on the discussion of the above two algorithms, it can be seen that PCC can reveal the linear relationship between meteorological features and bus load data, and DCC can respond to the non-linear relationship between meteorological features and bus load, and the combination of the two methods can enhance the rationality of meteorological feature selection. Therefore, this paper proposes to use PCC and DCC algorithms to calculate the absolute values of the correlation coefficients between meteorological features and bus load from linear and non-linear perspectives, respectively, and then sum them up to output the integrated correlation coefficients of each meteorological feature, and rank them in descending order, and select the features with the top 50% of the integrated correlation coefficient values as the input to the subsequent prediction models. In this paper, the integrated correlation coefficient is defined as:

$$\rho=\left|\rho_{PCC}\right|+\left|\rho_{DCC}\right| \tag{3}$$

where $\left|\rho_{PCC}\right|$ is the absolute value of the correlation coefficient calculated by the PCC method, $\left|\rho_{DCC}\right|$ and is the absolute value of the correlation coefficient calculated by the DCC method.

III. Forecasting model

A. Short term bus load prediction model based on GRU-Attention

The structure of the prediction model proposed in this paper is shown in Figure 1, including the input layer, the GRU layer, the dropout layer, the attention layer, and the output layer.

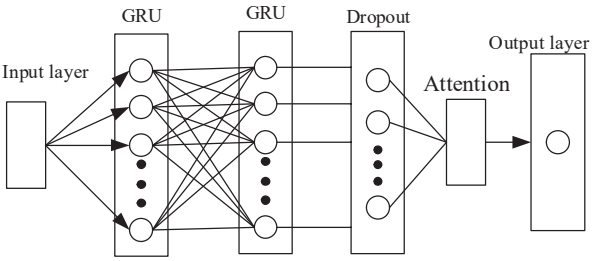


Figure 1. Structure of the GRU-Attention prediction model

(1) Input layer. Firstly, the original data is normalised, and the normalisation formula is shown in equation (4). The data after normalisation is used as input to the model, which is expressed as equation (5).

$$x^{'}=\frac{x-x_{\min}}{x_{\max}-x_{\min}} \tag{4}$$

where $x^{'}$ is the normalized value, x is the actual value, $x_{\max}$ and $x_{\min}$ are the maximum and minimum values in x .

$$X_u=\left[A_u,L_u\right] \tag{5}$$

Where X_u is the input vector at time t, A_u is the meteorological characteristic quantity at time u selected by the method in this paper, L_u is the bus load value corresponding to twelve consecutive moments before time u .

(2) GRU layer. Set two GRU layers to obtain state information from input data at different times.

(3) Droupout layer. In the process of neural network training, some neurons are discarded randomly to prevent overfitting and improve the generalization ability of the model.

(4) Attention layer. Using the output of the Dropout layer as input, and it can optimize the allocation of output weights for each hidden state.

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(5) Output layer. The bus load value at the next time is output through a fully connected layer.

B. Evaluation index

In order to evaluate the accuracy of bus load forecasting model, In this paper, root mean square error (RMSE), mean absolute error (MAE) and coefficient of determination (R^2) were used to evaluate the accuracy of the forecasting model. RMSE and MAE are used to measure the deviation between the forecasting value and the actual value. The closer the value is to 0, the more accurate the prediction. R^2 represents the degree of fit of the regression curve to the actual value, and the closer the value is to 1, the better the fit.

IV. EXAMPLE ANALYSI

A. Experimental data

In this paper, the actual bus load data sets of 220kV substation high voltage side in an area of our country and the weather information data sets in the area are used to conduct the experiments. The bus load data sets are selected power load data from June 1, 2019 to August 31, 2019, with a sampling interval of 15 minutes. The weather information data comes from the NASA Global Energy Resource Prediction Project and includes twelve meteorological factors such as temperature, wind speed and irradiance. As the sampling interval of the meteorological data is one hour, a linear interpolation method is used to match the bus load data.

B. Selection results of meteorological characteristics

In this paper, more than a dozen kinds of meteorological characteristic data and historical bus load data are taken as input of PCC and DCC algorithms. After calculation by the two algorithms, the correlation coefficients and comprehensive correlation coefficients of each characteristic quantity and bus load under different algorithms are shown in Table 1.

As can be seen from table 1, the comprehensive correlation coefficients of the all-day sunshine clarity index, the downward wave irradiance of the all-sky plane, the temperature at 2 meters, the wet bulb temperature at 2 meters, the short-wave down turning irradiance of the clear sky surface and the relative humidity at 2 meters are all higher than the comprehensive correlation coefficients of other weather characteristic quantities. Therefore, the above 6 weather characteristics are selected for forecast input in this paper.

TABLE 1. THE CORRELATION COEFFICIENT CALCULATED BY THE TWO ALGORITHMS

Characteristic quantity	$\lvert\rho_{PCC}\rvert$	$\lvert\rho_{DCC}\rvert$	ρ
The all-day sunshine clarity index	0.509	0.494	1.003
The downward wave irradiance of the all-sky plane	0.489	0.486	0.975
The temperature at 2 meters	0.455	0.430	0.855
The wet bulb temperature at 2 meters	0.441	0.418	0.859
The short-wave downturning irradiance of the clear sky surface	0.423	0.429	0.852
The relative humidity at 2 meters	0.406	0.399	0.805
The wind speed at 50 meters	0.361	0.310	0.671
The wind direction at 10 meters	0.270	0.285	0.555
The wind direction at 50 meters	0.268	0.286	0.554
The dew point at 2 meters	0.264	0.241	0.505
The wind speed at 10 meters	0.166	0.170	0.336
The surface pressure	0.047	0.191	0.238

C. Analysis of forecast results

1) Comparison of longitudinal experiments before and after the selection of meteorological characteristics

The forecast interval in this paper is 15 minutes, and the forecast objective of this paper is to forecast the bus load for 96 moments on August 31,2019. This paper compares the six weather features selected in Section 3.2 as model inputs with all features as model inputs. And the optimal parameters of the model obtained through several experiments are shown in Table 2, the results are shown in Figure 2, and the forecasting effect is shown in Table 3.

TABLE 2. PARAMETER SETTING OF GRU-ATTENTION

Optimization algorithm	Number of GRU layer	Number of neurons	drop rate of dropout layer
RMSProp	2	32/32	0.1

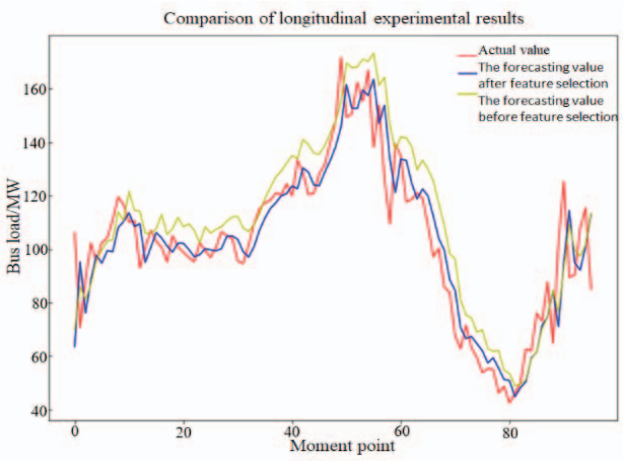


FIGURE 2 . FORECASTING RESULTS OF THE GRU-ATTENTION MODEL BEFORE AND AFTER FEATURE SELECTION

TABLE 3. COMPARISON OF LONGITUDINAL EXPERIMENTAL EVALUATION INDEX

Method	RMSE	MAE	R2
Feature selection	12.0989	8.8633	0.8193
Uncharacterized selection	14.7223	11.6717	0.7324

It can be seen from Figure 2 that compared with direct prediction without feature selection, the prediction curve obtained by the method proposed in this paper is closer to the

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actual bus load curve. Table 3 shows that compared with direct prediction, RMSE and MAE respectively decreased by 17.82% and 24.06%, and R2 increased by 11.87%. Through the above analysis, it can be seen that the method proposed in this paper can select appropriate meteorological characteristics as the input of the forecasting model, and has a high forecasting accuracy.

2) Transverse experimental comparison of different model predictions

In order to verify the effectiveness of the proposed method, a single LSTM model and a single GRU model are selected for comparison. The bus load forecasting results of different models at 96 moments on August 31, 2019 are shown in Figure 3, and the forecasting effect is shown in Table 4.

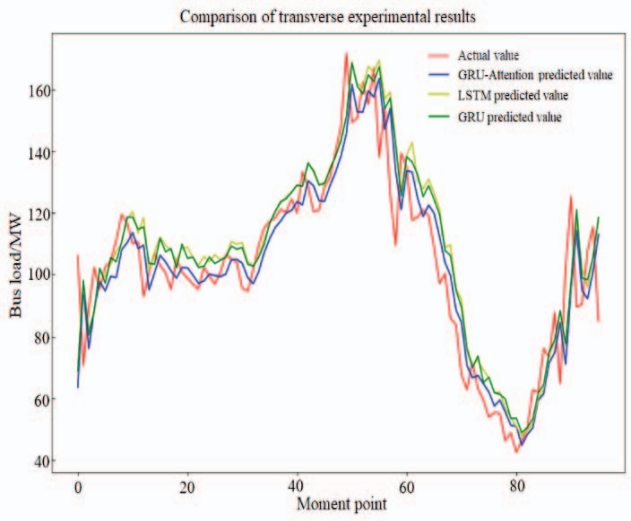


FIGURE 3. FORECASTING RESULTS OF DIFFERENT MODELS

TABLE 4. COMPARISON OF LONGITUDINAL EXPERIMENTAL EVALUATION INDEX

Method	RMSE	MAE	R2
GRU-Attention	12.0989	8.8633	0.8193
LSTM	13.6719	10.2677	0.7692
GRU	13.1902	9.9698	0.7852

As shown in Figure 3 and Table 4, after the Attention mechanism was added to the prediction model, RMSE respectively decreased by 11.51% and 8.27%, MAE respectively decreased by 13.68% and 11.10%, and R2 respectively increased by 6.51% and 4.34%, compared with a single LSTM model and a single GRU model. The proposed method has the highest degree of prediction curve fitting and the best error evaluation indexes, which proves the effectiveness and accuracy of the proposed method.

V CONCLUSION

Aiming at the problem that the weather characteristic quantity related to bus load is difficult to select due to different regions, this paper proposes a bus load prediction method based on double algorithm combined feature selection, and verifies the effectiveness of the proposed method through experiments. The conclusions are as follows: (1) the combined feature selection method can reasonably select the weather characteristics with high correlation with bus load and remove the low correlation, so as to improve the prediction accuracy of the model; (2) from the comparison results of transverse experiments, the proposed method has advantages in the three evaluation indexes, which can better restore the changing trend of bus load and has higher accuracy.

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REFERENCES

[1] Liu B, Qin C, Ping Ju, et al. Short-term bus load forecasting based on fusion of XGBoost and Stacking models [J]. Electric Power Automation Equipment,2020, 40(3):147-153.

[2] Wu D, Wang ZF. Day-ahead load forecasting based on day-by-day segmented meteorological algorithm[J]. Electrotechnical Application,2020,39(6):61-67.

[3] Huang Q. Research on probabilistic forecasting method of short-term load based on convolutional neural network[D].Nanning: Guangxi University,2021.

[4] Lu Y, Wang MK. Research on probabilistic forecasting method of short-term load based on convolutional neural network[J]. Automation of Electric Power Systems, 2006,30(16):83-85.

[5] Ding Q, Lu JG, Qian YM, et al. A practical extrapolation method for ultra - short - term load forecasting curve[J]. Automation of Electric Power Systems,2004,28(16):83-85.

[6] Jiang H, Zhang Y, Muljadi E, et al. A short-term and high-resolution distribution system load forecasting approach using support vector regression with hybrid parameters optimization[J]. IEEE Transactions on Smart Grid,2018,9(4):3341-3350.

[7] Chen YC, Zhang DH, Li YX, et al. Multi-objective optimization based on DWT-MOSMA-SVM for short term bus load forecasting[J]. Electric Power Construction, 2023, 44(03):49-55.

[8] Ding SY, Tan XL, Ye M, et al. Bus load forecasting considering large-scale distributed PV interconnection [J]. Renewable Energy Resources,2021,39(01):117-122.

[9] Hu RL, Chen YC, Zhang DH, et al. Bus load prediction method based on SSA-Bi-LSTM neural network[J]. Guangdong Electric Power, ,2022,35(02):19-26.

[10] Yan HW, Sheng CG. Short-term bus load forecasting based on hierarchical clustering method and extreme Learning machine [J]. Journal of Computer Applications,2018,38(08):2437-2441.

[11] Ma R, Gong RG, Yang HJ. Fluctuation Prediction Method for Equivalent Load of BUS Based on Information Granulation and Support Vector Machine[J]. Proceedings of the CSU-EPSA, 2019,31(04):25-31.

[12] Jin SH, Zhu X, Hao WG, et al. Forecasting of bus-bar net load based on PSR-ARIMA and SVR[J]. Power Demand Side Management,2018, 20(02): 20-24.

[13] Shi T, Mei F, Lu J, et al. Phase space reconstruction algorithm and deep learning-based very short-term bus load forecasting[J]. Energies, 2019 ,12(22):1-17.

[14] Liu JH, Li JC, Yang LY, et al. Short-term household load forecasting based on EMD-SLSTM[J]. Power System Protection and Control, 2019,47(06):40-47.

[15] Wang LM, Wu XH, Zhao TL, et al. A scheme for rolling statistical forecasting of PM2.5 concentrations based ondistance correlation coefficient and support vector regression[J]. Acta Scientiae Circumstantiae, 2017,37(04):1268-1276.

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